Lohit Gali

AI/ML Engineer
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Summary

- AI / ML Engineer with 3+ years of experience building **ML Models** at the intersection of **autonomous vehicles, infrastructure telemetry, and compliance workflows**
- Experienced in **real-time inference ML pipelines** built with **Python, PyTorch/TensorFlow, OpenCV** and **CLI-based deployments**
- Proven ability to transition proof-of-concepts into governed, **production-ready workflows** using real-time inference pipelines and **cloud-native infrastructure**
- Hands-on experience designing LLM-powered agents, **AI-assisted analysis workflows**, data processing and security controls across **AWS, Azure, and enterprise environments**
- Strong background in **cloud deployment, CI/CD, SQL, DevOps tooling, and applied ML** for infrastructure and large-scale systems

Skills

Languages & Libraries: Python, SQL, TensorFlow, OpenCV, PyTorch, YOLOv8, Detectron2 (Mask R-CNN), scikit-learn
Machine Learning: supervised & unsupervised learning, classification, regression, clustering, model validation, Image Augmentation.
AI & Agentic Systems: LLMs (OpenAI, Anthropic), Frameworks (LangChain, LlamaIndex, Hugging Face, BedRock), Prompt Engineering, Tool-Calling Agents, RAG, Vector Search
Analytics & Statistics: exploratory data analysis, hypothesis testing, feature engineering, statistical modeling
Databases & Data Management: SQL (MySQL, PostgreSQL), NoSQL (MongoDB, DynamoDB)
Scripting & Automation: PowerShell, Python, Bash, Power Automate
Cloud Platforms & Services: AWS, Azure, GCP (Cloud Functions)
CI/CD & DevOps: Jenkins, GitHub Actions, Azure DevOps, Docker, Kubernetes, Terraform, Apache Kafka

Professional Experience

Epik Solutions *Jun 2025 – Present*
AI / ML Engineer
Cybersecurity & Threat Intelligence

- Developed **ML models using Python** to analyze **high-volume telemetry event data** and **cybersecurity datasets** used by internal security teams in production workflows.
- **Translated business and operational requirements** into model objectives, evaluation metrics, and deployment strategies by working closely with **security, infrastructure, and analytics stakeholders**
- Prototyped and built computer vision and predictive **ML pipelines using PyTorch, TensorFlow, and OpenCV** to process **sensor and operational data**
- Performed **data cleaning, transformation, and feature extraction** to support downstream ML tasks
- Applied **time-series aggregation and anomaly detection** logic to identify **emerging trends** across security telemetry
- Built **automation scripts using Bash and PowerShell** to support CLI-based deployments, monitoring, and operational troubleshooting

- Supported **cloud-based production deployments** by validating application behavior, monitoring early usage signals and addressing issues discovered after release

Samay Systems LLC *July 2024 – Jun 2025*
AI / ML Engineer
Cloud & Compliance

- Designed and trained **supervised and unsupervised ML models** addressing **compliance, audits, risk analysis** and industry-specific analytic use cases.
- Built **Python services** that collect, validate, and summarize evidence data from **AWS, Azure, GitHub, and Jira and several other MCP-enabled APIs**, creating consistent inputs for downstream ML workflows
- Built **AI-assisted compliance workflows** combining LLM reasoning with structured enterprise data using **MCP-based tool calling and n8n-orchestrated agents**
- Implemented **RAG-style querying** over **historical compliance data** using **vector databases**, allowing customers to ask **natural-language questions** and receive contextual answers.
- Designed **complex relational schemas and SQL workflows** to support **multi-tenant**, multi-framework **compliance validation** at scale.
- **Deployed** services using **AWS cloud-native infrastructure, Docker, Kubernetes, and CI/CD pipelines**, ensuring reliability across multiple **production environments**

Monarch Tractor *Jun 2023 – Aug 2023*
ML Intern
Autonomous Vehicles

- Supported development of **computer-vision and ML pipelines** for **semi-autonomous agricultural vehicles** operating in **real-world environments**.
- Built **Python-based** preprocessing pipelines to **transform raw sensor and image data** into training-ready datasets for **detection and prediction tasks**.
- Built **object detection and segmentation models** using **PyTorch, TensorFlow, and OpenCV** to support **perception** and prediction use cases, validating outputs against operational constraints for autonomous and **transport-related systems**.
- **Tested models using CLI-driven workflows**, gaining hands-on experience with **operational ML systems**.
- Collaborated closely with engineers and stakeholders to **refine requirements** and **validate model outputs against field expectations** for production deployments.
- Deployed models to cloud-hosted production environments, gaining hands-on experience with performance monitoring and inference validation.

ZeMoSo Technologies *Jul 2021 – Aug 2022*
Associate Data Scientist
LabOps Automation

- Contributed to a **cloud-based scientific operations platform** designed to trace experiments, samples, and data from project inception to analysis.
- Built backend services and **ML-adjacent pipelines** that validated, enriched, and structured **experimental data**, enabling reliable downstream modeling and decision-making.
- Supported **Proof-of-Concept ML experimentation workflows** by automating dataset preparation, feature generation, and **experiment tracking**.
- Integrated **CI/CD pipelines** to validate models, enforce quality checks, and promote artifacts while maintaining auditability and security controls aligned with governance standards.

Education

Master's in Computer Science, Oklahoma State University, Stillwater, OK
Bachelor's in Computer Science, JNTU Hyderabad, India

Certifications and Coursework

Introduction to Artificial Intelligence — IIT Madras, Online
Practical Machine Learning with TensorFlow— Google, IIT Madras, Online
Machine Learning — KTH Royal Institute of Technology (Sweden), IIT Madras, Online
An Independent Study using Explainable AI — Oklahoma State University
Deep Learning Specialization — DeepLearning.AI, Coursera, Online